

The Trading Strategy — explained simply, in depth

What the strategy does and why, in plain words and worked examples

This is the full guide to the trading idea behind `Backtesting/engine.py`. It goes step by step with real numbers, but keeps the language simple. If you can picture a stretched rubber band snapping back, you can follow all of it.

The whole journey in one line: **work out what's normal** → **spot when a price gap is stretched too far** → **fade it** → **ride the snap-back** → **get out at the right moment**.

1. The big idea (in one minute)

We trade **spreads** — the *gap* between two related oil prices. For example, the price of oil for July delivery minus the price for August delivery. That gap might be **\$0.74**.

We do **not** bet on whether oil goes up or down. We only bet on whether that **gap returns to normal** after it gets stretched.

Picture the gap as a rubber band. Most of the time it rests at a comfortable length. Now and then a burst of buying or selling **stretches** it — too wide or too narrow. History shows it usually **snaps back**. So when we see a stretched gap, we bet on the snap-back. That's the whole strategy:

When the gap is unusually stretched, bet that it returns to normal.

Everything else is about doing that carefully — measuring "stretched" honestly, and getting in and out at sensible points.

Why bet on the gap, not on oil itself? Because we buy one month and sell another at the same time, so the big swings in the *overall* oil price cancel out. What's left is a small, steady, well-behaved relationship — far easier to predict than "will oil go up?"

2. What we actually trade

Seven gaps, all built from **WTI** (American oil) and **Brent** (global oil) futures, in three shapes:

- **Calendar gaps** — the same oil, two different months (e.g. July vs August WTI). A pure bet on whether near-term oil is getting tighter or looser than later oil.
- **Butterflies** — a three-month combination that checks whether the *middle* month is out of line with its neighbours. The most surgical, market-neutral of the three.
- **Brent vs WTI** — the gap between the two kinds of oil, driven by shipping and regional supply differences.

In every case we **buy one contract and sell another at the same time**, so we own no outright oil — only the gap. One contract is 1,000 barrels.

3. The key question — what is "fair value," and can we borrow Phase 2's model?

To know whether a gap is stretched, we need a reference: its **fair value** (its "normal" level). The project already built a clever fair-value model in **Phase 2** — a statistical model that predicts each spread from real-world causes (oil stockpiles, the US dollar, market fear, the season). The obvious question: **can we just reuse that here?**

We looked carefully, and the answer is **no** — for three solid reasons:

1. **Its ingredients aren't in this data.** The Phase 2 model needs fundamentals — inventory levels, refinery activity, the dollar, the fear gauge. The intraday feed we're trading contains **only oil prices** — none of those ingredients. So the model simply **can't be computed** here.
2. **Its number is out of date.** Phase 2's fair value for the WTI July–August gap is about **2.18**. That same gap is trading at about **0.74** today — the contracts have re-priced since. Forcing the old number on would scream "this gap is cheap by 1.4!" *forever* — a permanent false signal, not a real one.
3. **It barely moves intraday.** Those fundamentals update daily (or weekly), so over a few days the model is basically a flat line. It can't explain the minute-by-minute wiggles we're actually trading, which come from order flow, not from slow fundamentals.

So we estimate fair value from the gap's own recent behaviour — see the next section. This isn't a step *down* from Phase 2; it's the *right tool for the timeframe*. Phase 2 still contributes: its proven track record for each gap (how often that gap historically snapped back) is carried in as a **confidence score** on every trade. We just don't use its *price* as the reference, because it can't be one here.

4. Step one — work out what "normal" looks like

We look at the gap's recent history — the **last 24 readings, about six hours** of trading — and take their **average**. That average is our "normal" (the fair value) for right now. As new readings arrive, the average rolls forward, so "normal" keeps up with a slowly drifting gap.

We also measure how much the gap **normally wiggles** — its typical jiggle from reading to reading. Call one of those a "**wiggle**." A gap that normally jumps around a lot needs to move *a lot* to count as stretched; a calm gap needs only a little. Measuring in wiggles puts every gap — calm or jumpy — on the **same fair ruler**.

5. Step two — measure how stretched it is

We turn "how far from normal" into one number, counted in wiggles. Traders call it a **z-score**; you can read it as *"how many normal wiggles away from normal are we?"*

- Around **0** = sitting right at normal.
- **+2 or more** = unusually **high** — the gap is "**rich**" (too wide).
- **-2 or less** = unusually **low** — the gap is "**cheap**" (too narrow).

Worked example. Normal is **0.74**, one wiggle is **0.05**, and right now the gap is **0.84**. Distance from normal = **0.10**. In wiggles: $0.10 \div 0.05 = 2.0$. So the stretch reading is **+2.0** — stretched wide, right at our trigger.

6. Step three — the entry rule

Simple and strict: **fade the gap once it is stretched to 2 wiggles or more.**

- If it's **rich** (stretched wide, +2 or more) → **sell** the gap, betting it narrows back.
- If it's **cheap** (stretched narrow, -2 or less) → **buy** the gap, betting it widens back.

That's the only entry condition. We picked **2 wiggles** because it's a genuinely unusual stretch — unusual enough that the snap-back is a real edge, but common enough to give a decent number of trades. (Lower, like 1, trades more but on weaker, noisier signals; higher, like 3, waits for rare extremes.)

7. Step four — the exit rules

Once we're in, we leave for the **first** of four reasons:

1. **It worked — take profit.** The gap snapped back to normal (within a quarter of a wiggle of it). We take the win. We wait for the **full** snap-back, not just a sliver — we want the whole move.
2. **We were wrong — stop loss.** Instead of snapping back, the gap stretched **further against us**, to 3.5 wiggles. We accept we're wrong and cut it.
3. **It's taking too long — time stop.** If a trade hasn't resolved within **about 12 hours**, we stop waiting and close it. This avoids sitting in dead trades that drift nowhere.
4. **The day is ending — flatten.** We never hold through the overnight or weekend gap (too much can happen while the market is closed). We close out before the break.

The shape of the bet. We enter at 2 wiggles, aim for a snap-back to 0 (a **1.75-wiggle** reward) and cut at 3.5 (a **1.5-wiggle** risk). So the reward is a touch bigger than the risk, *and* mean-reversion wins more often than it loses — that combination is the edge.

8. Step five — how big is each trade?

One unit per trade. Always. Simple.

This is a deliberate choice. The point of this backtest is to answer "*does the signal work?*" — so we keep the size fixed and small, and report the **raw per-unit result**. One contract is 1,000 barrels, so a **\$1** move in a gap is worth **\$1,000**.

An honest note. An earlier version tried clever position-sizing (risking a fixed 1% of a \$250,000 account per trade). It technically worked, but it multiplied every position $\sim 100\times$, which blew the profit *and* the losses up into scary six-figure numbers that obscured whether the underlying signal was any good. So we stripped it back out. **Sizing is a separate decision** you layer on *after* you trust the signal — not part of testing the signal. Here we test the signal, cleanly, at one unit.

We keep a pretend \$250,000 only as a starting line for the running profit/loss chart.

9. A full trade, start to finish

1. **A gap gets stretched.** WTI July–August normally rests near 0.74 . A burst of buying pushes it to 0.84 — a stretch of about **+2.0 wiggles**, rich. ✅ It's past our trigger.
2. **We enter — SELL the gap at 0.84**, betting it falls back toward 0.74 .
3. **The confidence score** is set from Phase 2's track record for this gap (WTI calendars reverted ~75% of the time historically) plus how stretched it is — here, a solid ~80/100.
4. **It snaps back.** A few hours later the gap is back to 0.74 — within a quarter-wiggle of normal. That's our take-profit. **We close and pocket the difference** (the gap fell 0.10 , $\times \$1,000$ per unit per \$1 = a small win).

A **losing** trade differs only at the end: the gap pushes *up* to 3.5 wiggles instead, our stop fires, and we take a small, pre-planned loss. Or it stalls and the 12-hour time stop closes it near break-even.

10. How we keep score

For every trade we record: which gap, buy or sell, the in/out prices and times, why we exited, how long we held, the profit or loss, and the worst/best point it passed through.

From all the trades together, three numbers matter most:

- **Profit factor** — total dollars won $\div$ total dollars lost. Above **1.0** = making money; **3.0** = won three dollars for every one lost. The fairest single score, because a "win-often-lose-big" trick can fake a high win rate but **cannot** fake a high profit factor.
 - **Expectancy** — the average profit per trade. The real bottom line.
 - **Biggest dip (drawdown)** — the worst peak-to-valley fall in the running total; the "pain" measure.
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11. The cost check — does it survive trading fees?

By default we run **before costs** (the pure signal). But we can also charge **realistic trading costs** (the small fee on every buy and sell).

The honest finding: **the gross signal is solid, but it's a high-frequency, small-edge strategy, so realistic costs eat most of it** — net of a typical fee it thins to roughly break-even. That's not a failure; it's the truth about this kind of strategy: *the prediction works, but the moves are small enough that execution cost is the deciding factor*. The fix would be to trade only the highest-conviction setups (fewer, bigger), but we report the honest gross and net rather than dress it up.

12. The results (latest live data)

On the live company feed (~350 fifteen-minute bars, 12–18 June 2026):

Measure	Result	What it means
Trades	76	76 stretched-gap opportunities taken
Gross profit	+\$4,270	total per-unit dollars made (before costs)
Win rate	78%	78% of trades were profitable
Profit factor	3.03	won \$3.03 for every \$1 lost
Expectancy	+\$56 / trade	the average trade made \$56
Max drawdown	–\$1,450	worst dip along the way
Net (after a 1¢/leg fee)	~break-even	costs eat the small intraday edge

Per gap, most were positive (the butterflies and Brent–WTI strongest); one — Brent Sep–Oct — actually **lost** money, which we show plainly rather than hide. These are small, honest, per-unit numbers — the raw signal, not a leveraged book.

13. The honest limits

1. **Not much data yet.** A few days of trading. It's a working method, not a final verdict — feed it more days to really trust the numbers.
 2. **"Normal" is a simple average.** A single sharp spike can tug the average a little; a more robust measure is a possible refinement.
 3. **Costs decide it.** Gross looks good; net is roughly break-even. Real profitability needs either lower costs or pickier, higher-conviction trades.
 4. **Oil only.** The data is WTI and Brent crude — no diesel, petrol, or other products.
 5. **One-unit sizing.** This is the raw signal, not a sized portfolio; sizing and risk limits are a separate layer on top.
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14. Mini-glossary (plain words)

- **Spread / gap** — the price difference between two related contracts, traded as one bet.
 - **Rich / cheap** — a gap that's unusually wide (rich) or narrow (cheap) versus normal.
 - **Normal / fair value** — the gap's recent resting level (here, the average of recent readings).
 - **Wiggle** — the gap's typical jiggle; the unit we measure "stretch" in.
 - **Stretch / z-score** — how many wiggles away from normal we are right now.
 - **Fade** — to bet *against* a move (sell what just jumped up, buy what just dropped).
 - **Snap-back / mean reversion** — the tendency of a stretched gap to return to normal.
 - **Stop loss** — the pre-set point where we admit we're wrong and cut the trade.
 - **Take profit** — closing a winning trade once it has snapped back.
 - **Profit factor** — dollars won ÷ dollars lost (above 1 = profitable).
 - **Drawdown** — the worst drop from a high point to a later low; the "pain" measure.
 - **Gross vs net** — results before costs (gross) versus after costs (net).
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15. How to run it

```
python Backtesting/engine.py          # before costs (the pure signal)
python Backtesting/engine.py --slip 0.01 # charge a per-leg fee; reports net too
python Backtesting/engine.py --live     # keep re-running on the latest data
```

Each run writes the full trade-by-trade story to `out/trades_log.md`, a spreadsheet to `out/trades.csv`, and feeds the live dashboard. The plain-English overview of the whole project is in `../PROJECT_REPORT.md`; the engine's methodology is in `IDEATION.md`.